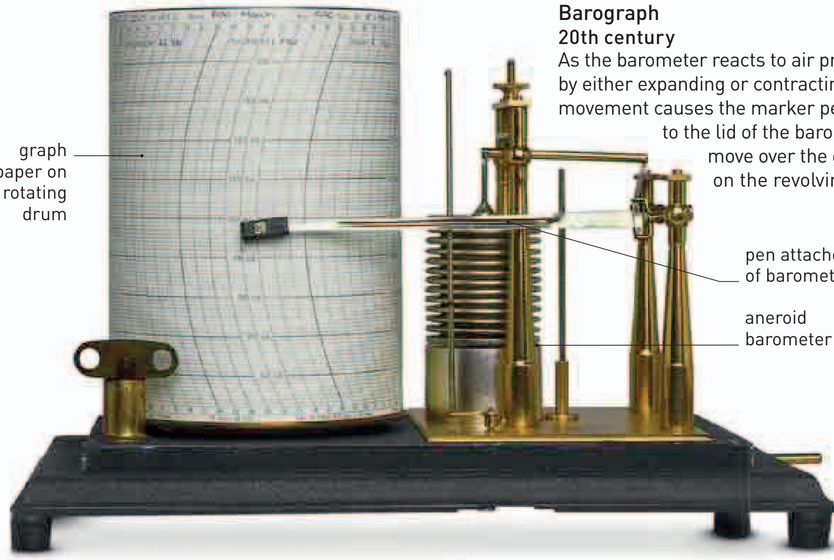


Barograph
20th century
As the barometer reacts to air pressure, by either expanding or contracting, the movement causes the marker pen attached to the lid of the barometer to move over the graph paper on the revolving drum.

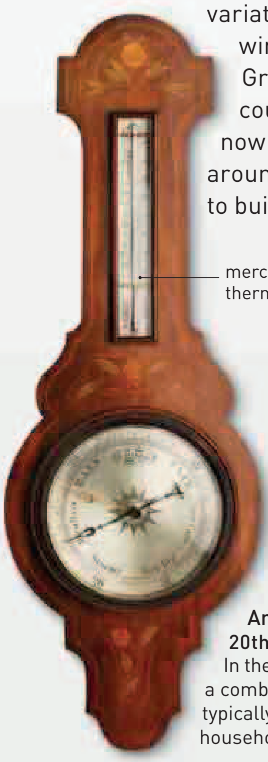
METEOROLOGICAL INSTRUMENTS

ACCURATE METEOROLOGICAL MEASUREMENTS PAVED THE WAY FOR WEATHER FORECASTING

People have always sought ways to understand and predict the changes to the weather around them, and this led to the development of devices to investigate the properties of air, such as its temperature and pressure.

The first meteorological instruments were made in 17th-century Italy. At first, tools were created simply to learn about the atmosphere. Thermometers measured changes in temperature; barometers revealed variations in air pressure; anemometers registered wind strength; and hygrometers showed humidity. Gradually, it was realized that these measurements could be used to help predict the weather, and now countless readings from weather stations all around the world are fed into powerful computers to build up weather forecasts.

Parheliometer
1881
Hours of sunshine can be recorded on a parheliometer, in which a glass ball focuses the Sun's rays onto card so that the Sun's passage leaves a scorch trace.



Aneroid barometer
20th century
Air pressure changes, shown on the dial of an aneroid barometer, are a good indicator of weather to come. Falling pressure suggests stormy weather, and steady high pressure heralds clear weather.

Spinning-cup anemograph
20th century
The spinning-cup anemometer was invented in 1846 by Irish astronomer John Robinson to measure wind speed. This "anemograph" records wind speed continuously on a cylindrical chart.

Aneroid barometer/thermometer
20th century
In the 19th century, before broadcast weather reports, a combined thermometer and aneroid barometer, typically housed in a case shaped like a banjo, helped householders make their own weather predictions.